Problem Summary

- Given the root of a binary tree, return the preorder traversal of its nodes' values

Preorder means root → left → right

Key Idea

Visit the current node first, then recursively visit:

→ left subtree

→ right subtree

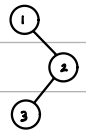
So the order is always:

→ current node

→ left child

→ right child

Diagram Examples



1 → 2 → 3

Approach

1) Create an empty result list

2) Define a helper function preorder(node)

3) If node is None, return

4) Add node value to result

5) Recurse on left subtree

6) Recurse on right subtree

7) Call preorder(root)

8) Return result

Time Complexity : O(n) → Each node is visited once

Space Complexity : O(h) → Recursion stack, where h is tree height + O(n) because result stores the values of all nodes